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ROTATABLE VEHICLE DISPLAY SYSTEM POWERED BY A MOTOR HAVING A FLEXIBLE CONNECTION TO THE DISPLAY SYSTEM

Abstract

The technology disclosed herein enables display screen rotation without compromising passenger impact safety. In a particular example, an apparatus includes a display system including a display screen on a front side of the display system and a mounting interface on a rear side of the display system. The apparatus also includes a bracket to which the mounting interface attaches and a flexible mechanical power transmitter connecting a power output of a motor to a power input at the mounting interface. The flexible mechanical power transmitter rotates the display system around a screen-rotation axis. The motor includes the power output, and the power output rotates around a motor-rotation axis different from the screen-rotation axis.

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Background/Summary

BACKGROUND

[0001] Center display screens in modern vehicles have evolved significantly over the years. These screens, often referred to as infotainment systems, serve as the central hub for various functions within the vehicle. From controlling climate settings and audio playback to providing navigation assistance, these displays enhance the driving experience. However, as these screens grow larger and more prominent in a vehicle's interior, the screens' involvement in passenger safety becomes an important factor to consider when designing a vehicle.

[0002] Government regulations have long provided safety standards that need to be met before a vehicle can be sold. Those safety standards include standards for passenger impact absorption by components of the vehicle's interior. Traditionally, those impact standards may describe how materials from which a vehicle's steering wheel or dashboard should be selected to mitigate the impact forces felt by a passenger should their head, or other body part, come into contact with the steering wheel or dashboard. In modern vehicles, the regulations may now apply to display screens, as passengers may also impact display screens when involved in an accident.

SUMMARY

[0003] The technology disclosed herein enables display screen rotation without compromising passenger impact safety. In a particular example, an apparatus includes a display system including a display screen on a front side of the display system and a mounting interface on a rear side of the display system. The apparatus also includes a bracket to which the mounting interface attaches and a flexible mechanical power transmitter connecting a power output of a motor to a power input at the mounting interface. The flexible mechanical power transmitter rotates the display system around a screen-rotation axis. The motor includes the power output, and the power output rotates around a motor-rotation axis different from the screen-rotation axis.

[0004] In another example, a method includes transmitting, from a vehicle compute unit, a rotation signal directing a motor to rotate the display system. The method also includes, at the motor, in response to the rotation signal, activating the motor connected to a flexible mechanical power transmitter to rotate the display system around a screen-rotation axis. Power output of the motor rotates around a motor-rotation axis different from the screen-rotation axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 illustrates a vehicle having a rotatable display system powered by a motor flexibly connected to the display system.

[0006] FIG. 2 illustrates implementation for rotating a display system using a motor flexibly connected to the display system.

[0007] FIG. 3 illustrates an operational scenario for a display system rotatable using a motor flexibly connected to the display system.

[0008] FIG. 4 illustrates implementation for rotating a display system using a motor flexibly connected to the display system.

[0009] FIG. 5 illustrates a chain drive for rotating a display system using a motor flexibly connected to the display system.

[0010] FIG. 6 illustrates an operational scenario for a display system rotatable using a motor flexibly connected to the display system.

[0011] FIG. 7 illustrates a mounting connection for rotating a display system using a motor flexibly connected to the display system.

[0012] FIG. **8** illustrates implementation for rotating a display system using a motor flexibly connected to the display system.

[0013] FIG. **9** illustrates an operation to rotate a display system using a motor flexibly connected to the display system.

[0014] FIG. **10** illustrates a computing system for rotating a display system using a motor flexibly connected to the display system.

DETAILED DESCRIPTION

[0015] Vehicle interiors are typically designed with safety in mind whether due to government regulations or general best practices. For instance, a dash and steering wheel of a vehicle may be designed in a manner to either absorb or, otherwise, reduce the force of a user impacting those parts of the vehicle. With larger displays becoming more prevalent, the chance of a user impacting a display is also greater. The display system described herein is affixed to a mounting bracket designed to collapse in a manner that absorbs impact forces (e.g., forces from a passenger's head or body during a front-end collision of the vehicle).

[0016] Moreover, the display apparatus reduces the impact force to a user while also providing a rotation mechanism for rotating the display. A rotation mechanism may be beneficial for displays of modern infotainment systems due to the new media types the systems are capable of presenting. For example, when presenting a movie in a widescreen format it may be preferable for the display to have a landscape orientation. In contrast, a portrait orientation may be preferable when driving and presenting a user interface into more typical vehicle functions (e.g., HVAC, radio, navigation, etc.). A rotation mechanism may rotate the display into either orientation to achieve the benefits of both orientations for a vehicle display. Importantly, the rotation mechanism described below does not sacrifice the impact absorption capabilities of the display system. A typical rigid drive shaft between motor and display would inhibit the display system from collapsing as designed. Similarly, including a rotation motor directly on the back of the display may add too much mass for the system to absorb an impact properly or the volume taken up by the motor may block the display system from collapsing in the desired manner.

[0017] FIG. **1** illustrates vehicle **100** having a rotatable display system powered by a motor flexibly connected to the display system. Vehicle **100** is depicted from the inside facing forwards. Vehicle **100** includes display system **101** as well as other common interior components (e.g., steering wheel, dash, passenger seats, gear selector, etc.). Display system **101** is rotatable 90 degrees to achieve either a portrait (solid line) or landscape (dashed line) orientation. Other examples may allow display system **101** to rotate more than the 90 degrees shown in this example. Vehicle **100** is merely an example design for a vehicle interior. As such, the components included in vehicle **100** may differ. Likewise, while shown as a passenger vehicle, vehicle **100** may be a car, truck, van, tractor, boat, airplane, wheelchair, or any other type of vehicle into which a display may be installed. Likewise, while illustrated as a center display in the front of vehicle **100**'s interior, display system **101** may be installed elsewhere in vehicle **100**, such as behind the front seats for viewing by rear passengers of vehicle **100**. A display system like display system **101** may also be useful in non-vehicular applications where impacts may occur to a rotating display system and absorbing those impacts would be beneficial. Bracket **205** described below would be mounted to a different structure in those applications than the vehicular examples presented herein.

[0018] FIG. **2** illustrates implementation **200** for rotating a display system using a motor flexibly connected to the display system. Implementation **200** includes display system **101**, motor **201**, power output **202**, flexible shaft **203**, cross-vehicle beam **204**, bracket **205**, and power input **206**, which are shown from a side profile. It should be understood that the shape of the components in implementation **200** is merely exemplary. Display system **101** includes display screen **221** and mounting interface **222**. Mounting interface **222** may be part of a housing for display screen **221** in display system **101**. Display screen **221** may be a Liquid Crystal Display (LCD), Light-Emitting Diode (LED) display, Organic Light-Emitting Diode (OLED) display, or some other type of display

technology. Mounting interface **222** attaches display system **101** to bracket **205**. Mounting interface **222** may be attached to bracket **205** using a threaded mechanism, tabs, clips, screws, nuts and bolts, rivets, or some other type of fastener-including combinations thereof. Mounting interface **222** may include one or more bearings enabling display system **101** to rotate around display rotation axis **211**, which runs perpendicular to the plane where display system **101** and bracket **205** meet. In some examples, the one or more bearings may be included in bracket **205** where mounting interface **222** connects to bracket **205**.

[0019] Motor **201** is an electric motor including power output **202**. Rotational power is provided by motor **201** via power output **202**, which rotates around motor rotation axis **212**. Motor **201** may be a direct current or alternating current motor. Motor **201** may include one or more bearings, rotor, stator, armature, magnets, brushes, or some other type of component for converting electrical power into mechanical power-including combinations thereof. Flexible shaft **203** is connected from power output **202** to power input **206**. Flexible shaft **203** may be a braided wire manufactured by braiding together multiple strands of wire, such as music wire, to produce a durable yet flexible mechanical power transmitter. Other examples may use different types of mechanical power transmitter, such as a chain or a belt, to transmit the rotation of power output **202** by motor **201** to power input **206**. While power output **202** is shown protruding from motor **201**, power output **202** in other examples may be flush with or recessed within the housing of motor **201**. Similarly, while power input **206** is not shown protruding from motor **201**, as power input **206** may be flush with or recessed into bracket **205** or mounting interface **222**, power input **206** may protrude in other examples such that power input **206** can be viewed in implementation **200**. Flexible shaft **203** may be connected to power output **202** and power input **206** using clamps at each end, pins or screws through power input **206** at each end, a keyed connector, or come other type of coupling that prevents power input **206** from releasing from power output **202** or power input **206**—including combinations thereof.

[0020] In this example, bracket **205** is affixed to cross-vehicle beam **204** and motor **201** is mounted to cross-vehicle beam **204**. Cross-vehicle beam **204** spans vehicle **100** from side to side. Cross-vehicle beam **204** may be located behind a dashboard of vehicle **100** when viewed from the passenger compartment of vehicle **100**. Other components of vehicle **100** may also be connected to cross-vehicle beam **204**. While motor **201** is directly mounted to cross-vehicle beam **204** in this example, motor **201** may be mounted to a bracket attached to cross-vehicle beam **204** or may be mounted elsewhere in vehicle **100** as long as motor **201** can still rotate display system **101** via flexible shaft **203**. Additionally, motor **201** is positioned such that motor **201** does not interfere with movement of display system **101** (at least for a predetermined distance) should an impact to display system **101** cause bracket **205** to deform allowing display system **101** to move towards cross-vehicle beam **204**. Display rotation axis **211** and motor rotation axis **212** are not parallel to each other in this example but may be parallel in other examples. In other examples, bracket **205** may be affixed to some other part of vehicle **100**'s structure or may be integrated into some other vehicle component rather than existing as an independent part. In those examples, vehicle **100** may not include cross-vehicle beam **204** or cross-vehicle beam **204** may not be positioned in a location conducive to attaching bracket **205**. In a particular example, bracket **205** may be integrated into a structure for vehicle **100**'s instrument panel.

[0021] While not shown, implementation **200** may also include one or more wires providing electrical power to display system **101** and enabling exchange of communications with display system **101** (e.g., to send a graphical display signal from a vehicle compute unit to display a user interface on display system **101** or receive user input via touchscreen input received from a passenger of vehicle **100**).

[0022] FIG. **3** illustrates operational scenario **300** for a display system rotatable using a motor flexibly connected to the display system. Operational scenario **300** is an example of what may happen with the components of implementation **200** should an impact force be applied to display

system **101**. The direction of the impact force on display system **101** is substantially towards cross-vehicle beam **204**, as the impact force may be coming from an impact of a passenger of vehicle **100** on display system **101**. For example, if vehicle **100** is in a front-end collision, the collision may cause a passenger's head to impact display system **101**. The chances of injury are likely greater if display system **101** did not move and cushion the impact in response to receiving the impact force. [0023] Bracket **205**, in this example, is designed to collapse, crush, or otherwise deform upon an impact force being transmitted to bracket **205** from display system **101**. The amount of force absorbed by display system **101** in conjunction with bracket **205** may be linear or progressive. The shape of bracket **205** may be dependent upon an amount of force intended to be absorbed, packaging considerations within vehicle **100**, a material (e.g., metal, plastic, etc.) from which bracket **205** is constructed, or some other characteristic. Given that motor **201** is not mounted to bracket **205**, bracket **205** may be a bracket originally designed to handle non-rotatable display systems. The ability to use existing bracket designs, which are already shown to comply with impact safety standards/regulations, for a rotating display system may result in cost savings for a manufacturer of vehicle **100** versus having to design a new bracket. Similarly, a manufacturer of vehicle **100** may be able to offer the option of installing a rotating display system or a static display system while using the same bracket for both.

[0024] In operational scenario **300**, bracket **205** has bent at multiple locations to absorb the impact force applied to display system **101**. Space still remains between display system **101** and motor **201** but the distance between display system **101** and motor **201** is smaller than the distance prior to impact, as shown in implementation **200**. The state of bracket **205** in operational scenario **300** is merely an example of how bracket **205** may crush in response to the impact force. As noted above, bracket **205** may take different shapes and be made of different materials, which may result in different reactions by bracket **205** to receiving the impact force. The flexibility of flexible shaft **203** enables flexible shaft **203** to bend when display system **101** moves with the collapsing of bracket **205** due to the impact force. Preferably, the flexibility of flexible shaft **203** is enough such that the presence of flexible shaft **203** has a negligible effect on impact absorption characteristics of bracket **205**. Likewise, motor **201** is positioned such that motor **201** does not interfere with movement of display system **101** or bracket **205** for at least a predefined amount of movement when an impact for is experienced by display system **101**. The amount of movement may be based on safety regulations or other passenger safety requirements. If flexible shaft **203** had been a rigid shaft (with motor **201** positioned to accommodate the rigid shaft), the rigid shaft may not yield to the impact force, which prevents display system **101** from collapsing as shown.

[0025] FIG. **4** illustrates implementation **400** for rotating a display system using a motor flexibly connected to the display system. Implementation **400** is like implementation **200** but shows an alternative form of flexible power transmission. Specifically, implementation **400** uses chain **403** as the flexible mechanical power transmitter instead of flexible shaft **203**. Chain **403** may be replaced by a belt in some examples. Implementation **400** includes display system **101**, motor **401**, power output **402**, chain **403**, cross-vehicle beam **404**, bracket **405**, power input **406**, which are shown from a side profile. It should be understood the shape of the components in implementation **400** is merely exemplary. Display system **101** includes display screen **421** and mounting interface **422**. Mounting interface **422** may be part of a housing for display screen **421** in display system **101**. Display screen **421** may be a Liquid Crystal Display (LCD), Light-Emitting Diode (LED) display, Organic Light-Emitting Diode (OLED) display, or some other type of display technology. Mounting interface **422** attaches display system **101** to bracket **405**. Mounting interface **422** may be attached to bracket **405** using a threaded mechanism, tabs, clips, screws, nuts and bolts, rivets, or some other type of fastener-including combinations thereof. Mounting interface **422** may include one or more bearings enabling display system **101** to rotate around display rotation axis **411**, which runs perpendicular to the plane where display system **101** and bracket **405** meet. In some examples, the one or more bearings may be included in bracket **405** where mounting interface **422** connects to

bracket **405**.

[0026] Motor **401** is an electric motor including power output **402**. Power output **402** is a shaft with a cog at one end over which chain **403** is installed. Rotational power is provided by motor **401** via power output **402**, which rotates around motor rotation axis **412**. Motor **401** may be a direct current or alternating current motor. Motor **401** may include one or more bearings, rotor, stator, armature, magnets, brushes, or some other type of component for converting electrical power into mechanical power-including combinations thereof. Power input **406**, like power output **402**, is a shaft with a cog at one end over which chain **403** is installed. When the cog on power output **402** is turned by motor **401**, chain **403** moves and turns the cog on power input **406**. power input **406** being connected to display system **101** causes display system **101** to rotate whenever power input **406** is rotated by chain **403**. Display rotation axis **411** and motor rotation axis **412** are substantially parallel with one another in this example to ensure chain **403** does not bind when moving. Similarly, the cogs of power input **406** and power output **402** are positioned substantially on the same plane to ensure chain **403** does not bind.

[0027] In this example, bracket **405** is affixed to cross-vehicle beam **404** and motor **401** is mounted to cross-vehicle beam **404**. Cross-vehicle beam **404** spans vehicle **100** from side to side. Cross-vehicle beam **404** may be located behind a dashboard of vehicle **100** when viewed from the passenger compartment of vehicle **100**. Other components of vehicle **100** may also be connected to cross-vehicle beam **404**. While motor **401** is directly mounted to cross-vehicle beam **404** in this example, motor **401** may be mounted to a bracket attached to cross-vehicle beam **404** or may be mounted elsewhere in vehicle **100** as long as power output **402** and power input **406** are oriented relative to one another such that chain **403** can move without binding. Additionally, motor **401** is positioned such that motor **401** does not interfere with movement of display system **101** (at least for a predetermined distance) should an impact to display system **101** cause bracket **405** to deform allowing display system **101** to move towards cross-vehicle beam **404**.

[0028] FIG. 5 illustrates chain drive **500** for rotating a display system using a motor flexibly connected to the display system. Chain drive **500** shows chain **403** from a different angle than implementation **400**. Specifically, chain **403** is shown from a perspective facing the rear of display system **101**. Chain drive **500** includes input cog **501** of power input **406** and output cog **502** of power output **402**. Input cog **501** and output cog **502** are the same size in this example, which results in a 1:1 drive ratio. In other examples, input cog **501** and output cog **502** may be different sizes to create different drive ratios. A distance between input cog **501** and output cog **502** may be chosen such that chain **403** can flex in response an impact force to display system **101**, as described in operational scenario **600** below. When power output **402** rotates output cog **502**, chain **403** moves as well. When chain **403** moves input cog **501** of power input **406** is moved to rotate display system **101**.

[0029] FIG. 6 illustrates operational scenario **600** for a display system rotatable using a motor flexibly connected to the display system. Operational scenario **300** is an example of what may happen with the components of implementation **400** should an impact force be applied to display system **101**. The direction of the impact force on display system **101** is substantially towards cross-vehicle beam **404**, as the impact force may be coming from an impact of a passenger of vehicle **100** on display system **101** (e.g., a passenger's head to impacting display system **101** during a front-end collision).

[0030] Bracket **405**, in this example, is designed to collapse, crush, or otherwise deform upon an impact force being transmitted to bracket **405** from display system **101**. The amount of force absorbed by display system **101** and bracket **405** may be linear or progressive. The shape of bracket **405** may be dependent upon an amount of force intended to be absorbed, packaging considerations within vehicle **100**, a material (e.g., metal, plastic, etc.) from which bracket **405** is constructed, or some other characteristic. Given that motor **401**, like motor **201**, is not mounted to bracket **405**, bracket **405** may be a bracket originally designed to handle non-rotatable display systems.

[0031] In operational scenario **600**, bracket **405** has bent at multiple locations to absorb the impact force applied to display system **101**. Space still remains between display system **101** and motor **401** but the distance between display system **101** and motor **401** is smaller than the distance prior to impact, as shown in implementation **400**. The state of bracket **405** in operational scenario **600** is merely an example of how bracket **405** may crush in response to the impact force. As noted above, bracket **405** may take different shapes and be made of different materials, which may result in different reactions by bracket **405** to receiving the impact force. The flexibility chain **403** enables chain **403** to bend when input cog **501** and output cog **502** become misaligned due to the collapsing of bracket **405** from the impact force. Preferably, the flexibility of chain **403** is enough such that the presence of chain **403** has a negligible effect on impact absorption characteristics of bracket **405**. In some examples, if chain **403** is flexed beyond a threshold amount of deflection, chain **403** may be designed to pop off of input cog **501** and/or output cog **502** or may break (e.g., a link in chain **403** may fail) before appreciably affecting the impact force absorption characteristics of bracket **405**. Likewise, motor **401** is positioned such that motor **401** does not interfere with movement of display system **101** or bracket **405** for at least a predefined amount of movement when an impact force is experienced by display system **101**. The amount of movement may be based on safety regulations or other passenger safety requirements. Like with the flexible shaft example above, the flexibility of chain **403** and the positioning of motor **401** allows bracket **405** to crumple as designed in contrast to what may occur with a rigid shaft or direct motor mounting.

[0032] FIG. 7 illustrates mounting connection **700** for rotating a display system using a motor flexibly connected to the display system. Mounting connection **700** includes mounting interface **701**, bracket face **705**, four mounting fasteners **702**, and power input **706**. Mounting interface **701** is an example of mounting interface **422** or mounting interface **222** for mounting display system **101**. Bracket face **705** is an example of a face of bracket **205** or bracket **405** to which mounting interface **222** or mounting interface **422** connects. As such, mounting connection **700** is shown from a perspective facing the rear of display system **101**. Bracket face **705** includes an opening through which mounting interface **701** can be seen and through which power input **706** passes. Power input **706** is an example of power input **206** or power input **406**. In some examples, a portion of mounting interface **701** may also pass through bracket face **705**. While the opening is round in this example, the opening in other examples may take different shapes.

[0033] Mounting interface **701** in this example includes four mounting fasteners **702** positioned around the opening. Mounting fasteners **702** may be bolts placed through holes in bracket face **705** and mounting interface **701** before being secured by nuts. Other examples may use more or fewer fasteners than the four shown in mounting connection **700**. In some examples mounting interface **701** may thread into the opening in bracket face **705**. Other manners of connecting mounting interface **701** to bracket face **705** may be used in other examples. Mounting interface **701** includes components, such as a bushing or bearing, that enables power input **706** to rotate along with the rest of display system **101** connected to power input **706**. In some examples, mounting interface **701** may include a component (e.g., a retaining pin) that prevents display system **101** from rotating when no mechanical power transmitter (e.g., shaft, chain, or belt drive) is connected to power input **706** or power input **706** is absent.

[0034] FIG. 8 illustrates implementation **800** for rotating a display system using a motor flexibly connected to the display system. Implementation **800** includes display system **801**, motor **802**, vehicle compute unit (VCU) **803**, and flexible mechanical power transmitter **813**. Display system **801** is an example of display system **101**, motor **802** is an example of motor **201** or motor **401**, and flexible mechanical power transmitter **813** is an example of flexible shaft **203** or chain **403**. Display system **801** and VCU **803** communicate over communication link **811**. Motor **802** and VCU **803** communicate over communication link **812**. Communication links **811-812** may be direct links or may include intervening systems, networks, or devices.

[0035] VCU **803** is a processor-based computing system that is at least tasked with controlling

rotation of display system **801**. VCU **803** may handle other vehicle tasks as well. VCU **803** includes a communication interface for communication links **811-812**. The communication interface may also handle other communication links with VCU **803**. VCU **803** may also include circuitry for connecting to sensors or other types of vehicle components (e.g., buttons, other motors, other displays, driver assistance systems, etc.). In some examples, VCU **803** may be one of multiple VCUs that each handle at least a portion of the tasks for vehicle **100**.

[0036] FIG. **9** illustrates operation **900** to rotate a display system using a motor flexibly connected to the display system. In operation **900**, VCU **803** receives a signal from display system **801** instructing VCU **803** to rotate display system **801** (step **901**). The signal may represent user input received from a user via display system **801**. For example, a software button displayed by display system **101** to give the user an option to rotate display system **801** may be triggered by the user and cause the signal to be sent to VCU **803**. The signal need not be received from display system **801** in all examples. In some examples, a physical button or switch, either on display system **801** or elsewhere in vehicle **100**, may be activated by the user. In further examples, the signal may include information received by VCU **803** about vehicle **100** and the information may trigger VCU **803** to rotate display system **801** automatically upon condition(s) being satisfied. For example, VCU **803** may receive a signal indicating vehicle **100** has been taken out of park (e.g., into drive or reverse). VCU **803** may be configured to ensure display system **801** is in a particular orientation when not in park and may, therefore, trigger rotation of display system **801** if vehicle **100** is not in the configured orientation.

[0037] After receiving the signal, VCU **803** determines whether rotation is allowed (step **902**). VCU **803** may maintain a set of conditions in which display system **801** cannot be in one orientation or the other, or conditions in which display system **801** should not be in motion regardless of orientation. Using the park example above, VCU **803** may be configured to maintain display system **801** in a portrait orientation when not in park. If the received signal is requesting a rotation into landscape while vehicle **100** is not in park, then VCU **803** will not perform the screen rotation. Optionally, VCU **803** may present an error message to inform the user that display system **801** cannot be rotated (step **903**). For instance, VCU **803** may send a signal back to display system **801** to display a message to the user and/or send a signal to an audio system of vehicle **100** to play an audible alert to the user. In some examples, rotation may be allowed in all circumstances and step **902** can be skipped.

[0038] If screen rotation is allowed, VCU **803** sends a motor activation signal to motor **802** (step **904**) and motor **802** activates to rotate display system **801** in response to the signal (step **905**). Motor **802** may include processing circuitry to process and carryout instructions from VCU **803**. For instance, the signal may direct motor **802** to rotate and the circuitry in motor **802** may carry out that instruction on its own until display system **801** is in the desired orientation. In another example, the signal may turn on motor **802** and VCU **803** may send another signal directing motor **802** to stop (e.g., a sensor on display system **801** may detect when the desired orientation is reached and provide a feedback signal to VCU **803** triggering VCU **803** to stop motor **802**). In other examples, communication link **812** may be a power supply line to motor **802**. In those examples, the signal from VCU **803** comprises power to motor **802**, which causes motor **802** to rotate. A physical stopper or sensor at display system **801** may indicate to VCU **803** when display system **801** has reached the desired orientation.

[0039] FIG. **10** illustrates computing system **1000** for rotating a display system using a motor flexibly connected to the display system. Computing system **1000** is an example architecture for VCU **803**. Computing system **1000** includes storage system **1045**, processing system **1050**, and communication interface **1060**. Processing system **1050** is operatively linked to communication interface **1060** and storage system **1045**. Communication interface **1060** may be communicatively linked to storage system **1045** in some implementations. Computing system **1000** may further include other components such as a battery and enclosure that are not shown for clarity.

[0040] Communication interface **1060** comprises components that communicate over communication links, such as network cards, ports, radio frequency (RF), processing circuitry and software, or some other communication devices. Communication interface **1060** may be configured to communicate over metallic, wireless, or optical links. Communication interface **1060** may be configured to use Time Division Multiplex (TDM), Internet Protocol (IP), Ethernet, optical networking, wireless protocols, communication signaling, or some other communication format—including combinations thereof.

[0041] Processing system **1050** comprises microprocessor and other circuitry that retrieves and executes operating software from storage system **1045**. Storage system **1045** may include volatile and nonvolatile, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Storage system **1045** may be implemented as a single storage device but may also be implemented across multiple storage devices or sub-systems. Storage system **1045** may comprise additional elements, such as a controller to read operating software from the storage systems. Examples of storage media include random access memory, read only memory, magnetic disks, optical disks, and flash memory, as well as any combination or variation thereof, or any other type of storage media. In some implementations, the storage media may be a non-transitory storage media. In some instances, at least a portion of the storage media may be transitory. In no examples would storage media of storage system **1045**, or any other computer-readable storage medium herein, be considered a transitory form of signal transmission (often referred to as “signals per se”), such as a propagating electrical or electromagnetic signal or carrier wave.

[0042] Processing system **1050** is typically mounted on a circuit board that may also hold the storage system. The operating software of storage system **1045** comprises computer programs, firmware, or some other form of machine-readable program instructions. The operating software of storage system **1045** comprises screen rotator **1030**. The operating software on storage system **1045** may further include an operating system, utilities, drivers, network interfaces, applications, or some other type of software. When read and executed by processing system **1050** the operating software on storage system **1045** directs computing system **1000** to control a motor that rotates a display system. Screen rotator **1030** may execute natively on processing system **1050** or the operating software may include virtualization software, such as a hypervisor, to virtualize computing hardware on which screen rotator **1030** executes.

[0043] In an example, screen rotator **1030** directs processing system **1050** to receive a signal instructing processing system to rotate a display system and transmit a rotation signal directing a motor to rotate the display system. In response to the rotation signal, the motor, which is connected to a flexible mechanical power transmitter, rotates the display system around a screen-rotation axis. Power output of the motor rotates around a motor-rotation axis different from the screen-rotation axis.

[0044] The included descriptions and figures depict specific implementations to teach those skilled in the art how to make and use the best mode. For teaching inventive principles, some conventional aspects have been simplified or omitted. Those skilled in the art will appreciate variations from these implementations that fall within the scope of the invention. Those skilled in the art will also appreciate that the features described above can be combined in various ways to form multiple implementations. As a result, the invention is not limited to the specific implementations described above, but only by the claims and their equivalents.

Claims

1. An apparatus for a motorized rotation of a display in a vehicle, the apparatus comprising: a display system including a display screen on a front side of the display system and a mounting interface on a rear side of the display system; a bracket to which the mounting interface attaches; a

flexible mechanical power transmitter connecting a power output of a motor to a power input at the mounting interface, wherein the flexible mechanical power transmitter rotates the display system around a screen-rotation axis; and the motor including the power output, wherein the power output rotates around a motor-rotation axis different from the screen-rotation axis.

2. The apparatus of claim 1, wherein the flexible mechanical power transmitter comprises a flex shaft.
3. The apparatus of claim 2, wherein the flex shaft comprises braided wire.
4. The apparatus of claim 1, wherein the power output comprises an output gear and the power input comprises an input gear, and wherein the flexible mechanical power transmitter comprises a chain engaging the output gear and the input gear.
5. The apparatus of claim 4, wherein the output gear rotates around a motor-rotation axis substantially parallel to the screen-rotation axis.
6. The apparatus of claim 1, wherein the power output comprises an output pulley and the power input comprises an input pulley, and wherein the flexible mechanical power transmitter comprises a belt engaging the output pulley and the input pulley.
7. The apparatus of claim 1, wherein the motor is mounted to a cross-vehicle beam within a dash of the vehicle.
8. The apparatus of claim 7, wherein the bracket is mounted to the cross-vehicle beam.
9. The apparatus of claim 1, wherein the bracket is configured to buckle when the display system is impacted from the front side while still providing clearance between the display system and the motor.
10. The apparatus of claim 1, wherein the flexible mechanical power transmitter deforms from an operating position when the display system is impacted from the front side.
11. The apparatus of claim 1, wherein the motorized rotation is limited to rotating the display between a portrait orientation and a landscape orientation.
12. A method for rotating a display system in a vehicle, the method comprising: transmitting, from a vehicle compute unit, a rotation signal directing a motor to rotate the display system; and at the motor, in response to the rotation signal, activating the motor connected to a flexible mechanical power transmitter to rotate the display system around a screen-rotation axis, wherein power output of the motor rotates around a motor-rotation axis different from the screen-rotation axis.
13. The method of claim 12, comprising: receiving, in the vehicle compute unit, user input requesting the display system be rotated, wherein the rotation signal is transmitted in response to the user input.
14. The method of claim 12, wherein the rotation signal comprises electrical power to power the motor.
15. The method of claim 12, wherein the rotation signal comprises an instruction to power the motor.
16. The method of claim 12, wherein activating the motor rotates the display system between a portrait orientation and a landscape orientation.
17. An apparatus for a motorized rotation of a display in a vehicle, the apparatus comprising: a display mounted to a bracket; a cross-vehicle beam; the bracket attached to the cross-vehicle beam; and an interface between the bracket and the display, wherein the interface allows the display to rotate when connected to a flexible mechanical power transmitter connecting a power output of a motor to the display.
18. The apparatus of claim 17, comprising: the motor attached to the cross-vehicle beam; and the flexible mechanical power transmitter.
19. The apparatus of claim 17, wherein the interface prevents the display from rotating when not connected to a flexible mechanical power transmitter.
20. The apparatus of claim 17, wherein the bracket is designed to collapse upon receiving a specified impact force.

